

Boot Menu in IGEL OS 12

During the boot process, a boot menu is available on request. Through this menu, you can start boot modes for troubleshooting. There are modes to access system parameters, or to reset the device to the factory defaults if the device is configured incorrectly or if you experience problems when booting.

→ During the boot process, press the [Esc] key repeatedly in rapid succession in the second stage loader, when the `loading kernel` message is shown on the screen.
The boot menu is displayed with the available boot modes:



Using the arrow keys, navigate to one of the boot modes and press the [Enter] key to start the process. You can start the following boot modes:

- [Quiet boot](#): Normal startup. (Default)
- [Verbose boot](#): Start with system messages and an interactive root shell
- [VESA only boot](#): Basic graphic boot
- [Emergency boot \(setup only\)](#): Only the **Setup** window is available
- [Failsafe boot with CRC check](#): Start with an integrity check of the operating system
- [Reset to factory defaults](#): Reset the client to factory defaults
- [Custom boot command](#): Boot with configurable command line options

Quiet Boot

Quiet boot is the default boot mode. It is the normal startup mode. In this mode, all kernel messages are disabled and the graphical user interface is started.

Verbose Boot

Unlike in **Quiet boot** mode, the kernel messages are shown in **Verbose boot** mode. The boot process also pauses before the graphics system and the user session start.

This gives you an opportunity to open a root shell and interactively execute debugging commands (for example, `ifconfig`).

Only use the root shell if you have adequate knowledge of Linux or if you are instructed to do so by the IGEL Helpdesk and are given appropriate guidance. Incorrect use can destroy the operating system.

To execute debugging commands:

1. Select **Verbose boot** from the boot menu.
2. Wait until the boot messages stop at Reached target IGEL Network Online.
3. Open a virtual console with one of the key combinations:
 - [Ctrl] + [Alt] + [F11]
 - [Ctrl] + [Alt] + [F12]
4. Log in by pressing [Return] and enter the root password if necessary.
5. Go through the desired individual commands.
6. Now enter the following command to continue the normal boot process: `systemctl default`
The graphical user session starts.

VESA Only Boot

Use this boot mode if normal boot has graphic issues, for example, if the device has limited Graphical Processor Unit (GPU) support. This mode is not manufacturer specific. In this mode resolution and multimonitor mode and performance might be limited.

Emergency Boot (Setup Only)

In the **Emergency boot** mode, the device is started without network drivers and with a resolution of 640 x 480 - 60 Hz. After the boot process, the **Setup** window is opened automatically.

This mode is useful, for example, if you have selected an excessively high screen resolution or a wrong mouse type and these settings can no longer be changed in the normal setup. Unlike with a reset, the setup opens with the actual settings.

→ Once you are done with the changes, close the setup window to reboot the device.

Failsafe Boot with CRC Check

During a **Failsafe boot**, a check of the file system is carried out first. Then, the **Verbose boot** is started.

This mode is helpful if you no longer have a bootable system after a firmware update. The **Failsafe boot** checks where the problem is. If need be, an old version will be booted and you will need to repeat the firmware update.

Reset to Factory Defaults

If you select **Reset to factory defaults**, all personal settings on the device (including your password and the sessions you have configured) will be lost.

You can reset IGEL OS devices to factory defaults through the boot menu, but the best practice is to reset devices using the OSC as described in [How to Deploy IGEL OS 12 with IGEL OS Creator \(OSC\)](#).

You can also reset your device to factory defaults through the UMS Web App. In this case, the device will be removed from the UMS and you will have to register your device with the UMS again. For details, see [How to Reset a Device to Factory Defaults via the IGEL UMS Web App](#).

Before the reset procedure is carried out:

1. Confirm the displayed warning message.
2. If the device is protected by an administrator password, enter the password. You have three attempts.

The administrator password can be set or changed under [Security > Password > Administrator](#) using a profile or the device configurator in the IGEL UMS Web App.

Custom Boot Command

In the **Custom boot command** mode, preconfigured options are placed on the kernel command line. This allows you, for example, to investigate and rectify problems with specific hardware components.

The **Custom boot command** is merely a temporary solution – it is not an everyday booting method. It must therefore be selected manually in the boot menu.

To configure the options for the **Custom boot command**, proceed as follows:

1. Open a local terminal and log in as `root`.
2. Enter the following command to bring up the current options:
`bootreg get /dev/igfdisk boot_cmd`
3. Save your desired options with the following command:
`bootreg set /dev/igfdisk boot_cmd "<Your Options>"`
4. Check the options that you have entered:
`bootreg get /dev/igfdisk boot_cmd`

If you would like to delete options for the Custom boot command, leave an empty string of characters in their place: `bootreg set /dev/igfdisk boot_cmd ""`

IGEL Tech Video

In the video, IGEL OS11 is used for demonstration.